

Analysis PhD Exam, May 2002

I

State the following theorems:

1. The Egorov theorem.
2. The Radon–Nikodym theorem.
3. Density of step functions in L^p and L^∞ .
4. The Fubini theorem for $\mu \times \nu$ -integrable functions.
5. The Riesz representation theorem of the dual of $C(K)$.

II

State and prove one of the following theorems:

1. The integral representation of the dual of $L^1(\mu)$ (the finite measure case).
2. The theorem of integration with respect to $g\mu$.

III

Solve the following problems:

1. Let (X, Σ, μ) be a measure space with $\mu(X) < \infty$, F a Banach space, and $\mathcal{F} \subseteq \Sigma$ a σ -algebra. Prove the existence of the conditional expectation $E(f \mid \mathcal{F})$ for $f \in L_F^1(\mu)$.
2. Let $(X, \mathcal{S})$, $(Y, \mathcal{T})$ be measurable spaces and $\mu : \mathcal{S} \rightarrow \mathbb{R}$, $\nu : \mathcal{T} \rightarrow \mathbb{R}$ σ -additive measures. Prove the existence of the product measure $\mu \times \nu$ and the equality $\|\mu \times \nu\| = \|\mu\| \times \|\nu\|$.
3. Let $(X, \mathcal{S})$, $(Y, \mathcal{T})$ be measurable spaces, $p : X \rightarrow Y$ an $(\mathcal{S}, \mathcal{T})$ -measurable mapping, $\mu : \mathcal{S} \rightarrow \mathbb{R}_+$ a σ -additive measure, and $\nu = p(\mu)$ the image measure of μ under the mapping p . Prove that if $f \in L_F^1(\nu)$, then $f \circ p \in L_F^1(\mu)$ and

$$\int f \circ p \, d\mu = \int f \, d\nu.$$

4. Let (X, Σ, μ) be a measure space and $f \in L^1(\mu)$. Prove that for every $\varepsilon > 0$, there is a set $X_\varepsilon \in \Sigma$ with $\mu(X_\varepsilon) < \infty$ such that

$$\int_{X \setminus X_\varepsilon} |f| \, d\mu < \varepsilon.$$

5. Let λ be the Lebesgue measure on $[0, 1]$ and $f \in L^1(\lambda)$. Prove that if

$$\int_{[0,1]} x^n f \, d\lambda = 0$$

for every n , then $f = 0$, λ -a.e.

Hint: Use the fact that the continuous functions on $[0, 1]$ are dense in $L^1(\lambda)$ and reduce to the case $\int_A f \, d\lambda = 0$ for $A \in \mathcal{B}([0, 1])$.

Note. Write complete computations and explanations. Do not use arrows or other unnecessary signs. Use words for explanations. Write complete sentences. State the complete framework for each theorem.